

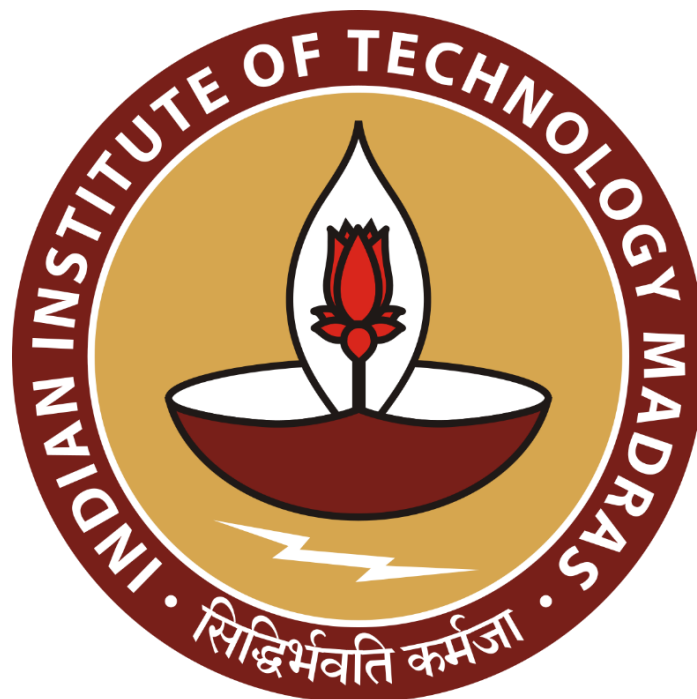
Demand Forecasting and Revenue Optimization for an E-Commerce Marketplace

A Proposal report for the BDM capstone Project

Submitted by

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Contents

Declaration Statement	3
1 Executive Summary	4
2 Organization Background	4
3 Problem Statement	5
3.1 Investigating Demand Fluctuations and Seasonal Revenue Patterns	5
3.2 Identifying Drivers of Delivery Delays and Order Cancellations	5
3.3 Customer Segmentation for Retention and Marketing Optimization	5
4 Background of the Problem	5
5 Problem Solving Approach	6
5.1 Methods and Analytical Techniques	6
5.2 Data Source and Repository Details	7
5.3 Analysis Tools and Software	7
6 Expected Timeline	8
6.1 Work Breakdown Structure	8
6.2 Gantt Chart	8
7 Expected Outcome	9

Declaration Statement

I am working on a Project titled “Demand Forecasting and Revenue Optimization for an E-Commerce Marketplace”. I extend my appreciation to Kaggle and the Olist data team for making the publicly available structured dataset accessible for academic and research purposes.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through a reputable secondary data repository and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project’s completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfilment in the BS Degree Programme offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate:

A handwritten signature in black ink, appearing to read 'Nishika Chapra', written over a horizontal line.

Name: Nishika Chapra

Date: April 13th, 2026

1 Executive Summary

The organization at the center of this study is an established Brazilian e-commerce marketplace that connects thousands of small and medium-sized merchants with end consumers across all 26 states of Brazil. Operating under a Business-to-Consumer (B2C) model, the platform facilitates multi-category online retail transactions including electronics, home goods, fashion, and personal care, while also managing logistics, payment processing, and post-sale customer support. The dataset used in this project is a publicly available structured secondary dataset sourced from the Kaggle repository and represents real transactional records spanning from 2016 to 2018.

The principal business challenges identified for this study center on three interrelated operational inefficiencies. First, the platform experiences significant seasonal demand fluctuations that create mismatches between available inventory, active sellers, and consumer purchase volumes, negatively impacting revenue consistency. Second, a measurable proportion of orders are subject to delivery delays beyond the estimated fulfilment date, resulting in negative customer reviews and reduced repeat purchase rates. Third, the absence of structured customer segmentation means that marketing resources are allocated broadly without insight into high-value or at-risk customer cohorts.

To address these challenges, this project applies a multi-method data-driven analytical framework. Exploratory Data Analysis (EDA) establishes baseline patterns; ARIMA and Prophet-based time-series models forecast future demand cycles; K-Means clustering segments the customer base; and regression modelling identifies the strongest predictors of delivery delay and revenue performance. The expected outcomes include an operational demand forecast model, an evidence-based customer segmentation framework, and actionable recommendations to reduce order cancellations and improve marketing effectiveness, all validated through a Streamlit-based interactive dashboard.

2 Organization Background

The organization under study is a large-scale Brazilian e-commerce marketplace founded in 2015 and headquartered in São Paulo, Brazil. It operates as a B2C intermediary platform that connects independent sellers and small-to-medium enterprises (SMEs) with consumers across the country, functioning in a manner analogous to a digital shopping mall. The platform's revenue model is

commission-based: sellers list their products across diverse categories – including electronics, furniture, beauty products, sporting goods, and toys – while the platform earns a percentage of each completed transaction.

By 2018, the marketplace had onboarded approximately 3,000 unique sellers operating across 27 Brazilian states, serving a registered customer base exceeding 96,000 individuals. The platform processed over 99,000 orders during the period covered by this dataset, generating a total gross merchandise value (GMV) in the range of BRL 13.5 million. The organization manages its own last-mile logistics infrastructure through partnerships with multiple freight carriers, enabling nationwide delivery coverage.

Given its position as a high-growth platform in one of Latin America's largest e-commerce markets, the organization faces mounting pressure to optimize operational performance, reduce delivery failures, and improve customer lifetime value through data-informed decision-making.

3 Problem Statement

3.1 Investigating Demand Fluctuations and Seasonal Revenue Patterns: The marketplace experiences inconsistent order volumes across months and product categories. This objective aims to identify seasonal demand trends and their revenue impact.

3.2 Identifying Drivers of Delivery Delays and Order Cancellations: A significant share of orders experience fulfilment delays beyond the estimated delivery date, lowering customer satisfaction. This objective seeks to quantify and model the key contributing factors.

3.3 Customer Segmentation for Retention and Marketing Optimization: Without structured segmentation, the platform cannot identify high-value or at-risk customers. This objective uses clustering to create actionable buyer personas to guide targeted marketing and retention strategies.

4 Background of the Problem

The Brazilian e-commerce sector has grown at an accelerated pace over the past decade, driven by widespread smartphone adoption, expanding internet penetration, and the rise of digital payment platforms. Despite this growth, marketplace operators face structural inefficiencies that erode profitability and customer loyalty if left unaddressed.

Internally, the platform's key challenges arise from a decentralized seller ecosystem. With thousands of independent merchants managing their own inventory and fulfilment processes, the platform has

limited direct control over stock availability, packaging quality, and shipping timelines. This results in an unpredictable supply-side performance that amplifies the impact of sudden demand spikes. Additionally, the absence of a data-driven demand forecasting system means the platform cannot proactively communicate expected volume surges to sellers during peak seasons such as the Brazilian Black Friday (November) or national holidays.

Externally, the platform operates across a geographically diverse country with significant logistical disparities between urban centers (São Paulo, Rio de Janeiro) and remote northern or north-eastern states. Road infrastructure limitations, regional carrier capacity constraints, and customs clearance for cross-state shipments contribute to delivery delay variability. Furthermore, consumer behavior on the platform exhibits sensitivity to review scores: orders with historical delivery failures in a given product category correlate with reduced conversion rates.

Taken together, these internal and external pressures make it essential for the organization to develop a robust, analytics-driven understanding of demand patterns, delivery performance, and customer behavior – forming the core motivation for this capstone project.

5 Problem Solving Approach

5.1 Methods and Analytical Techniques

The project adopts a structured five-stage analytical pipeline grounded in the BDM curriculum framework:

1. **Exploratory Data Analysis (EDA):** Univariate and bivariate statistical summaries will be computed across all nine dataset files. Visualizations including histograms, correlation heatmaps, and time-line plots will be used to identify distributions, outliers, and inter-variable relationships relevant to the three problem objectives.
2. **Time-Series Demand Forecasting:** Monthly order volume and revenue will be modelled using ARIMA (Autoregressive Integrated Moving Average) and Facebook's Prophet, which handles seasonal decomposition natively. These models will produce 6-month forward demand forecasts to support inventory and seller readiness planning.
3. **K-Means Clustering (Customer Segmentation):** RFM (Recency, Frequency, Monetary) features will be engineered from the orders and payments datasets. K-Means clustering will then be applied to group customers into actionable segments (e.g., loyal, at-risk, dormant). Optimal k will be determined via the Elbow Method and Silhouette Score.
4. **Regression Modelling (Delivery Delay Prediction):** A multiple linear regression model will be constructed to quantify the effect of order weight, freight value, seller state, customer state, and order timestamp on delivery delta (actual vs. estimated delivery date). Feature importance will guide targeted operational improvements.
5. **Review Score Analysis:** The order reviews dataset will be cross-referenced with delivery performance to assess whether delivery delays are the primary driver of low review scores, providing a closed-loop metric for the organization.

5.2 Data Source and Repository Details

This project uses the Olist Brazilian E-Commerce Public Dataset, a structured, multi-table secondary dataset publicly available on Kaggle. The dataset comprises nine interrelated CSV files covering 99,441 orders placed between October 2016 and August 2018, and includes variables spanning order status, payment type, product category, seller location, freight value, customer geography, delivery timestamps, and consumer review scores.

Dataset source URL: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Supplementary marketing funnel dataset: <https://www.kaggle.com/datasets/olistbr/marketing-funnel-by-olist>

This dataset was selected because it (a) represents real-world B2C marketplace operations rather than synthetic data, (b) contains sufficient volume (~100K records) for statistically robust modelling, (c) spans multiple years enabling genuine time-series analysis, and (d) directly supports all three problem objectives through its comprehensive coverage of orders, logistics, payments, and customer feedback.

5.3 Analysis Tools and Software

Python (version 3.10+) is the primary analysis environment, utilizing the following libraries:

- Pandas & NumPy – data wrangling, missing-value treatment, feature engineering
- Matplotlib & Seaborn – static visualization (distributions, heatmaps, time-series plots)
- Plotly & Folium – interactive charts and geospatial heatmaps for regional analysis
- Scikit-learn – K-Means clustering, regression modelling, cross-validation
- Statsmodels / ARIMA – classical time-series decomposition and forecasting
- Prophet (Facebook) – seasonal decomposition and demand forecasting with holiday effects
- Streamlit – interactive dashboard presenting all major findings and forecasts
- Google Colab / Jupyter Notebook – reproducible analysis notebooks for submission

All code, notebooks, and the Streamlit dashboard will be hosted on a public GitHub repository with a comprehensive README, enabling full reproducibility and serving as a professional portfolio artefact.

6 Expected Timeline

6.1 Work Breakdown Structure

Figure 1 below presents the hierarchical decomposition of all project tasks across three primary phases: (i) Data Collection and Preprocessing, (ii) Data Analysis and Modelling, and (iii) Documentation and Reporting.

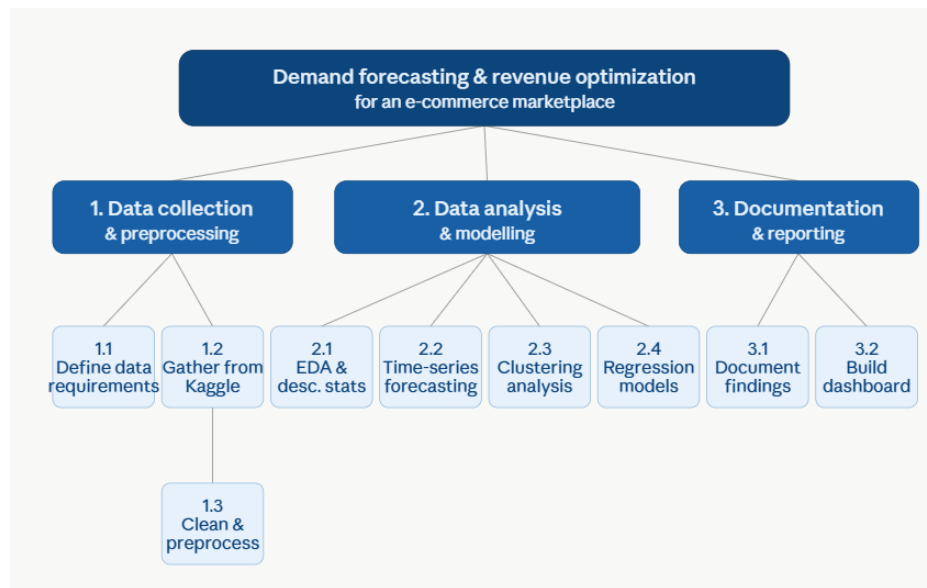


Figure 1: Work Breakdown Structure

6.2 Gantt Chart

Figure 2 presents the project timeline over an approximate 24-day working schedule, indicating task durations, dependencies, and parallel workstreams.

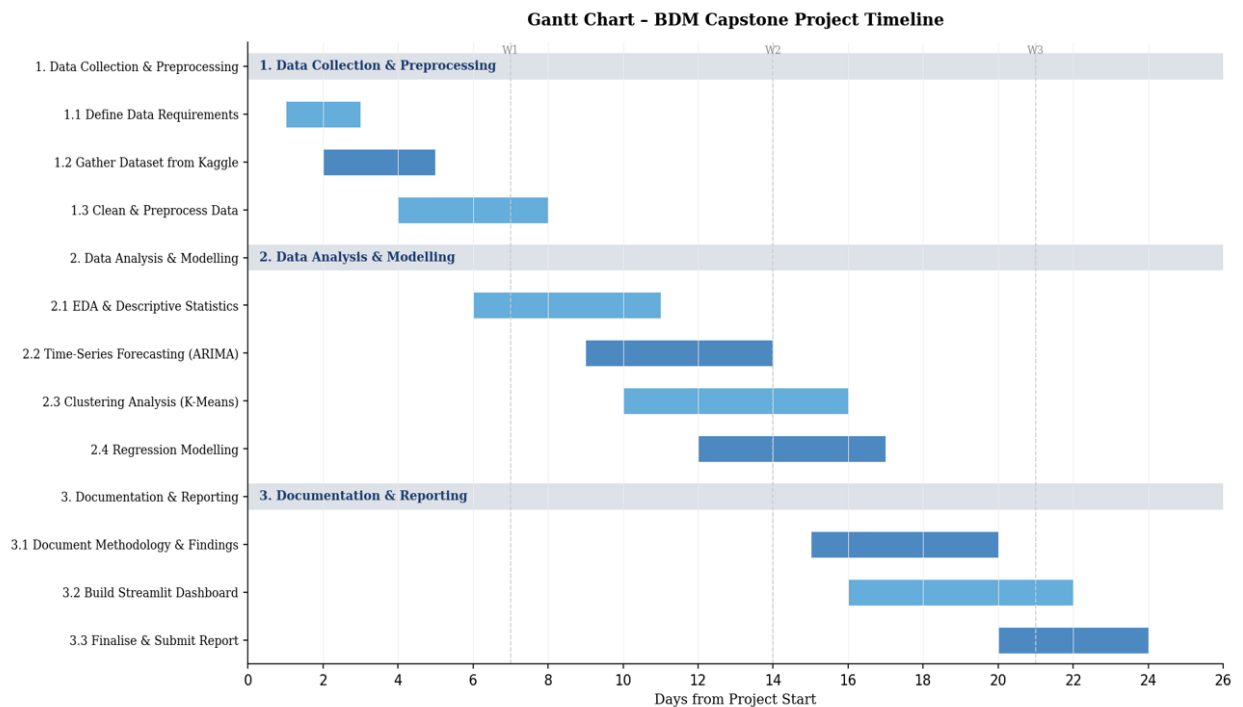


Figure 2: Gantt Chart showing expected timeline for project completion

7 Expected Outcome

The completion of this project is expected to deliver the following concrete, data-driven outcomes for the e-commerce marketplace:

- **Demand Forecasting Model:** A validated ARIMA/Prophet model producing monthly order volume and revenue forecasts, enabling proactive seller and logistics capacity planning ahead of peak demand cycles.
- **Delivery Delay Prediction:** A regression model identifying the top predictors of delivery delays (e.g., seller state, freight route, product weight), with actionable recommendations to reduce the proportion of late deliveries.
- **Customer Segmentation Framework:** RFM-based K-Means clusters defining distinct buyer personas (e.g., high-value loyalists, one-time buyers, dormant users), directly informing targeted retention campaigns.
- **Interactive Business Dashboard:** A Streamlit-based dashboard synthesizing all key findings into an accessible decision-support tool for non-technical business stakeholders.

All insights will be exclusively derived from rigorous statistical and machine learning analysis of the dataset, ensuring they are evidence-based and replicable. The project output will be hosted on GitHub as a portfolio-grade resource.

References

1. **Olist Brazilian E-Commerce Dataset – Kaggle:**
<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>
2. **Olist Marketing Funnel Dataset – Kaggle:**
<https://www.kaggle.com/datasets/olistbr/marketing-funnel-by-olist>
3. **Facebook Prophet Documentation:** <https://facebook.github.io/prophet/>
4. **Scikit-learn Documentation:** <https://scikit-learn.org/stable/>
5. **IBM – Business Problem to AI/Data Science Solution:**
<https://www.ibm.com/garage/method/practices/discover/business-problem-to-ai-data-science-solution/>
6. **Project Manager – Executive Summary Guide:**
<https://www.projectmanager.com/blog/write-an-executive-summary>